



Academic Details			
Year	Degree	Institute	CGPA/Marks(%)
2026	B.Tech Computational Engineering	IIT Hyderabad	7.94
2022	XII (CBSE)	Saraswati Devi Public School	97%
2020	X (CBSE)	Saraswati Devi Public School	93.6%

Scholastic Achievements

WBJEE (West Bengal Joint Entrance Examination)

Achieved **Merit Rank 5127 & PWD Rank 11** in WBJEE out of 81393 shortlisted candidates. 2022

Positions of Responsibility

- Class Representative 2024-25** Aug. 2024 - Jun. 2025
 - Elected as CR of Computational Engineering batch'22
- Coordinator** Aug. 2024 - Jun. 2025
 - Selected as a coordinator under Design & PR domain of Extra Mural Lectures (EML) club.
 - Done relevant work under EML club.

Projects

Regression Model from scratch

- A regression model which is built from scratch using only numpy and matplotlib libraries of python.
- Linear regression, understanding model order and overfitting using regularization, effects of Lagrange parameter.
- Non-linear regression using various kernels, use of stochastic gradient descent for weight update, understanding bias-variance trade off, understanding MAP estimation.

Neural Network from scratch

- A neural network classification model that uses MNIST data set for digit recognition using three-layer feed-forward neural network.
- Using various activation functions like Sigmoid, Tanh, ReLu, Softmax functions along with optimized algorithms such as backpropagation, Stochastic Gradient Descent (SGD) to update weights, attaining 93.8 test accuracy.
- It also uses Adam optimization algorithm, thereby attaining accuracy up to 96%.
- To reduce the overfitting problem, it uses algorithms like weight regularization, dropout strategy, and early stopping.

Dynamic Matrix squaring using various mutual exclusion algorithms

- It involves performing parallel matrix multiplication using a dynamic mechanism in C++. The goal is to compute the square of a matrix A in parallel, with dynamic allocation of rows to threads.
- It includes the implementation of different mutual exclusion algorithms like Test-and-Set (TAS), Compare-And-Swap (CAS), Bounded Compare-And-Swap (Bounded CAS), Atomic increment (provided by the C++ atomic library) to handle synchronization issues.

Skills

Programming Languages :

- Proficient in:** C/C++, Python
- Familiar with:** SQL, Julia

Frameworks :

- Numpy, Pandas, Scipy, Matplotlib, Pywhatkit,

Tools :

- Solid Edge

Operating Systems :

- Windows, Linux

Relevant Courses

- Computer Science:** Introduction to Programming, Data Structures & Applications, Operating Systems.
- AI and Scientific Computing:** Foundation of Machine Learning, Fundamentals of Scientific Computing, Computer Aided Numerical Methods, Computational Methods in Material Science.
- Electrical Engineering:** Digital Circuits, Introduction to Hardware Description Language.
- Mathematics:** Calculus I & II, Series of Functions, Transform Techniques, Differential Equations, Linear Algebra, Matrix Theory, Probability & Statistics.

Extracurricular

- Participated in the Group Singing event of **Stranger Bands** organized by the Vibes Club.